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Cab and cargo body configuration for a delivery vehicle

Abstract

A vehicle includes a chassis and a vehicle body coupled to the chassis. The vehicle body includes a cab disposed on a forward end of the chassis and a cargo body disposed on a rear end of the chassis. The cab includes a front door that is slidably engaged to the cab. The cargo body includes a side door that is slidably engaged to the cargo body. The front door and the side door are disposed on the same side of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 16/426,656, filed May 30, 2019, which claims the benefit of and priority to U.S. Provisional Patent Application No. 62/678,735, filed May 31, 2018, the contents of which are hereby incorporated by reference in their entireties.

BACKGROUND
(1) Mail and package delivery vehicles transport a variety of cargo from post-office and shipping centers to residences and businesses. These delivery vehicles are configured to carry large quantities of material to reduce the number of trips between the post-office and various municipalities. The delivery vehicles are also configured so that the packages may be quickly loaded at the post-office or shipping center and are readily accessible to an operator during delivery. During a typical route, an operator may need to access mail boxes from within the vehicle to drop off smaller mail items, enter and exit the vehicle repeatedly to drop off larger packages, and access the cargo area to quickly locate packages for drop-off.
SUMMARY
(2) An embodiment relates to a vehicle. The vehicle includes a chassis and a vehicle body coupled to the chassis. The vehicle body includes a cab disposed on a forward end of the chassis and a cargo body disposed on a rear end of the chassis. The cab includes a front door that is slidably engaged to the cab. The cargo body includes a side door that is slidably engaged to the cargo body. The front door and the side door are disposed on the same side of the vehicle.
(3) In any of the above embodiments, a height of the cargo body may be greater than a height of the cab. In these instances, the vehicle may further include a roof cab assembly disposed proximate to an upper surface of the cab. The roof cap assembly may extend between an A-pillar and a B-pillar of the cab. An upper edge of the roof cap assembly may be approximately flush with an upper wall of the cargo body. Additionally, the roof cap assembly may be coupled to at least one of the upper surface or a forward surface of the cargo body. In some embodiments, the cab may include a cutout disposed in a top wall of the cab and the roof cap assembly may completely cover the cutout.
(4) Another embodiment relates to a vehicle. The vehicle includes a chassis and a vehicle body coupled to the chassis. The vehicle body includes a cab disposed on a forward end of the chassis and a cargo body disposed on a rear end of the chassis. The cab includes a front door that is slidably engaged to the cab. The cargo body includes a plurality of side walls, an upper wall, a forward partition, and a lower wall that together define a storage volume. The cargo body additionally includes a side door, an access door, and a rear door. The side door is slidably engaged to one of the plurality of side walls. The access door is slidably engaged to the forward partition. The rear door is slidably engaged to a rear facing one of the plurality of side walls. Each of the side door, the access door, and the rear door separate the storage volume from an environment surrounding the vehicle.
(5) In some embodiments, the front door and the side door are disposed on the same side of the vehicle. In some implementations, a height of the cargo body is greater than a height of the cab. In these instances, the vehicle may further include a roof cap assembly extending between an A-pillar and a B-pillar of the cab. An upper edge of the roof cap assembly may be approximately flush with the upper wall. Additionally, the roof cap assembly may be coupled to at least one of the upper surface of the cab or the forward partition.
(6) Another embodiment relates to a cargo body. The cargo body includes a plurality of side walls, an upper wall, a forward partition, and a lower wall that together define a storage volume. The

cargo body additionally includes a side door, an access door, and a rear door. The side door may be slidably engaged to one of the plurality of side walls. The access door may be slidably engaged to the forward partition. The rear door may be slidably engaged to a rear facing one of the plurality of side walls. Each of the side door, the access door, and the rear door separate the storage volume from an environment surrounding the vehicle.
(7) In some embodiments, the forward partition includes an access that is sized to accommodate a standing height of an occupant and the access door is disposed in the access. In some embodiments, the cargo body further includes a roof cap assembly. In these instances, an upper edge of the roof cap assembly may be approximately flush with the upper wall. Additionally, the roof cap assembly may be coupled to the forward partition.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:
(2) FIGS. **1-2** are perspective views of a mail and package delivery vehicle, according to an illustrative embodiment.
(3) FIG. **3** is a right side view of the mail and package delivery vehicle of FIGS. **1-2**.
(4) FIG. **4** is a left side view of the mail and package delivery vehicle of FIGS. **1-2**.
(5) FIG. **5** is a top view of the mail and package delivery vehicle of FIGS. **1-2**.
(6) FIG. **6** is a rear view of the mail and package delivery vehicle of FIGS. **1-2**.
(7) FIG. **7** is a perspective view of a cargo box for the mail and package delivery vehicle of FIGS. **1-2**.
(8) FIG. **8** is a left side view of the mail and package delivery vehicle of FIGS. **1-2**, at a cross-section parallel to a longitudinal axis of the vehicle.
(9) FIG. **9** is a perspective view of a track assembly for a side access door of the mail and package delivery vehicle of FIGS. **1-2**.
(10) FIGS. **10-11** are perspective views of a side access door for a mail and package delivery vehicle, according to an illustrative embodiment.
(11) FIG. **12** is a perspective view of the mail and package delivery vehicle of FIGS. **1-2** shown without a roof cap assembly.
(12) FIGS. **13-14** are perspective views of a roof cap assembly for a mail and package delivery vehicle, according to an illustrative embodiment.
(13) FIG. **15** is a side view of the roof cap assembly of FIGS. **13-14**.
(14) FIGS. **16-17** are rear and front views, respectively, of a cargo box of a mail and package delivery vehicle, according to an illustrative embodiment.
(15) FIG. **18** is a rear view of the cargo box of FIGS. **16-17** with the partition door installed.
(16) FIG. **19** is a perspective view of a track assembly for the partition door of FIG. **18**.
(17) FIG. **20** is a perspective view of a fully assembled cargo box for a mail and package delivery vehicle, shown with an upper wall of the cargo box removed, according to an illustrative embodiment.
(18) FIG. **21** is a top view of the cargo box of FIG. **20**.
(19) FIG. **22** is a top perspective cross-sectional view of the cargo box of FIG. **20**
(20) FIG. **23** is a perspective view of a shelf arrangement for the cargo box of FIG. **20**.
(21) FIG. **24** is an enlarged perspective view of a coupling mechanism for the shelf arrangement of FIG. **23**.
(22) FIG. **25** is an enlarged perspective view of a leveling bumper mechanism for the shelf

arrangement of FIG. **23**.

DETAILED DESCRIPTION

(23) Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

(24) As shown in FIGS. **1-6**, a mail and delivery vehicle includes a frame and a vehicle body mounted to the frame. The vehicle body further includes a cab toward the front of the delivery vehicle that is used to accommodate one or more vehicle operators. In an embodiment, the cab is a third-party designed vehicle cab (e.g., for a commercial utility vehicle, truck, etc.). The cab includes a sliding door that provides access to and from an interior volume of the cab, while minimizing the overall footprint of the delivery vehicle.

(25) In various embodiments, the cab of the delivery vehicle includes an opening along its upper wall that allows an occupant to stand at least partially erect (e.g., in a standing position) in the cab (e.g., that permits a user to raise his/her head above the upper wall of the cab). To protect the occupant and cab from exposure to the surrounding environment, the delivery vehicle includes a roof cap assembly that that couples the upper surface of the cab to a forward wall of the cargo body. An occupant is provided with three different access points to the cargo body. A first access point is an access door disposed in a partition between the cab and the cargo body. This access door is coupled to a track assembly in the partition and can be quickly opened by an occupant by sliding the access door along the partition wall. A second access point is a side access door disposed in a side wall of the cargo body. The side access door is similar to the sliding door on the cab and is located on the same side of the vehicle (e.g., passenger or driver side) as the sliding door. The side access door also includes a track assembly, which is configured to facilitate delivery operations while maximizing available storage volume at the interior of the cargo body. Finally, a third access point is a retractable (e.g., coiling door, rolling sheet, etc.) rear access door that provides larger clearance for oversized packages and delivery truck loading operations.

(26) The interior space of the cargo body is outfitted with multiple shelves that are uniquely configured to maximize the utility of the storage space. The shelves are disposed between adjacent frame rails of the cargo body and are connected to the frame rails. Advantageously, each shelf may be configured in either a folded position, where the shelves are retracted into the space between frame rails, or an extended position in which the shelf is turned downward to accommodate mailings and other packages. The arrangement of shelves may be modified to more efficiently utilize the interior space of the cargo body. The details of the general depiction provided in FIGS. **1-6** will be more fully explained by reference to FIGS. **7-25**.

(27) According to the exemplary embodiment of FIGS. **1-6**, a vehicle, shown as delivery vehicle **10** includes a chassis, shown as frame **20**, and a vehicle body, shown as body **30** that is rigidly coupled to the frame **20**. The body **30** of the delivery vehicle **10** includes a cab **40** coupled to a front end of the frame **20** and cargo body, shown as cargo box **50**, coupled to a rear end of the frame **20**. As shown in FIG. **3**, a height **51** of the cargo box **50** (e.g., up and down as shown in FIG. **3**), from an upper edge of the cargo box **50** to a lower edge of the cargo box **50** is greater than a maximum height **41** of the cab **40**. The cab **40** is configured to accommodate one or more occupants of the delivery vehicle **10**. In the delivery vehicle **10** of FIGS. **1-6**, the cab **40** is an enclosure that protects an occupant from environmental and road hazards. The cab **40** may include various components to facilitate operation of the delivery vehicle **10** by an operator. For example, the cab **40** may include an instrument panel that provides a user with diagnostic information about the delivery vehicle **10**. The cab may also include control equipment (e.g., steering wheel, throttle and brake pedals, signals, etc.) associated with the delivery vehicle **10**, which are contained within an interior space of the cab **40** for ready access by an operator. The delivery vehicle **10** also includes a prime mover, or engine, coupled to the frame **20** (e.g., at a position beneath the cab **40**). The engine is configured

to provide power to a plurality of tractive elements, shown as wheels **60**, and/or other systems of the delivery vehicle **10** (e.g., a pneumatic system, a hydraulic system, etc.). The engine may be configured to utilize one or more of a variety of fuels (e.g., gasoline, diesel, bio-diesel, ethanol, natural gas, etc.), according to various embodiments. The engine may additionally or alternatively include one or more electric motors coupled to the frame **20** (e.g., a hybrid refuse vehicle, an electric refuse vehicle, etc.). The electric motors may consume electrical power from an on-board storage device (e.g., batteries, ultra-capacitors, etc.), from an on-board generator (e.g., an internal combustion engine, etc.), and/or from an external power source (e.g., overhead power lines, etc.) and provide power to the systems of the delivery vehicle **10**.
(28) In various embodiments, the cab **40** of the delivery vehicle **10** includes a sliding door system, shown as sliding front door system **64** that facilitates user entry and exit from the delivery vehicle **10**. The sliding front door system **64** includes at least one front sliding door. As shown in FIGS. **1-2**, the front sliding door may be one of a left sliding door **66** slidably engaged to a left side (e.g., driver's side) of the cab **40** of the delivery vehicle **10**, or a right sliding door **68** slidably engaged to a right side (e.g., passenger's side) of the cab **40** of the delivery vehicle **10**. Each of the left sliding door **66** and the right sliding door **68** are structured to move (e.g., slide) relative to the delivery vehicle **10** in substantially parallel orientation with a side (e.g., a lateral side, a passenger or driver side) of the delivery vehicle **10**. As shown in FIGS. **1-2**, the sliding front door system **64** includes both a left sliding door **66** and a right sliding door **68**. The sliding front door system **64** further includes a plurality of track assemblies that are coupled (e.g., mounted) to at least one of the frame **20** and the cab **40**. In the illustrative embodiment of FIGS. **1-2**, the arrangement of track assemblies is similar for each of the left sliding door **66** and the right sliding door **68**. Among other benefits, incorporating sliding doors on the cab **40** improves accessibility into and out of the cab **40** as compared with hinged-type doors, particularly in tight spaces where available clearance in a lateral direction (into and out of the page of FIGS. **3-4**) away from the cab **40** is small.
(29) As shown in FIG. **7**, the cargo box **50** is a separate structure from the cab **40** (see also FIGS. **1-2**) and is coupled to the cab **40**. In other embodiments, the cab **40** and the cargo box **50** may be a single unitary structure. The cargo box **50** includes a plurality of side walls **52**, an upper wall **54**, a lower wall **56**, and a forward partition **100** that together define a storage volume configured to accommodate packages and other cargo. The side walls **52** and the forward partition **100** extend upwardly from the lower wall **56** in substantially perpendicular orientation with respect to the lower wall **56**. As shown in FIGS. **1-2**, the forward partition **100** is disposed proximate to the cab **40** and separates occupants in the cab **40** from the storage volume (e.g., separates an interior volume of the cab **40** from the storage volume). As shown in FIG. **7**, the cargo box **50** includes multiple access points that may be used by an occupant to access the storage volume. A first access, shown as access **110** is a rectangular cutout disposed centrally in the forward partition **100**. A second access, shown as side access **200** is a rectangular cutout disposed in a side wall **52** of the cargo box **50**. In an illustrative embodiment, the side access **200** is disposed just forward of a set of rear wheel wells **62** in the cargo box **50**. A third and final access, shown as rear access **300** is a rectangular cutout disposed in a rear facing side wall **52** of the cargo box **50**. As shown in FIG. **7**, the rear access **300** provides an occupant with greater clearance for accessing the storage volume as compared with access **110** or side access **200**.
(30) As shown in FIG. **8**, a side access door assembly **112** is disposed in side access **200** and prevents unwanted entry through the side access **200**. As shown in FIG. **3**, the side access door assembly **112** is spaced apart from (e.g., separated from) the sliding front door system **64** in a longitudinal direction (e.g., front to back of the delivery vehicle **10**) to accommodate the track assemblies for the sliding front door system **64**. As shown in FIGS. **10-11**, the side access door assembly **112** includes a rectangular side door **114** and a track assembly, shown as side track assembly **116** that is coupled to the side door **114**. The rectangular side door **114** is slidably engaged to the cargo box **50** via the side track assembly **116**. Among other benefits, using sliding

doors on both the cab **40** and the cargo box **50** (see FIG. **3**), such that there is two sliding doors on the same side of the delivery vehicle **10**, improves accessibility in tight spaces and reduces the overall footprint of the delivery vehicle **10** when the sliding doors are open (e.g., relative to hinged-type doors).

(31) The side track assembly **116** includes an upper track member **118** slidably coupled to the side door **114** proximate to an upper edge **117** of the rectangular side door **114**. As shown in FIG. **9**, the upper track member **118** is rigidly coupled to (e.g., rigidly mounted to, secured in position relative to) a surface of a first internal cavity of the side wall **52** that is proximate to the upper wall **54**. As shown in FIGS. **10-11**, the side track assembly **116** additionally includes a lower track member **120** slidably coupled to a lower edge **119** of the side door **114** and a central track member **122** slidably coupled to the side door **114** at a central position between the upper edge **117** and the lower edge **119**. Similar to the mounting configuration for the upper track member **118** shown in FIG. **9**, the lower track member **120** is rigidly coupled to a surface of a second internal cavity of the side wall **52** proximate the lower wall **56** of the cab **40**. As shown in FIGS. **1** and **3**, the central track member **122** is rigidly coupled to the side wall **52** of the cab **40** in a central position between the upper track member **118** and the lower track members **120**.

(32) Each of the upper track member **118**, the lower track member **120**, and the central track member **122** is arranged to extend in a substantially longitudinal direction (e.g., front-to-back of the delivery vehicle **10**) to facilitate movement of the side door **114** between a closed position, where the side door **114** fully occludes the side access **200** (as shown in FIG. **3**), and an open position, where at least a portion of the side door **114** is positioned at least partially rearward of the side access **200**. Each of the track members **118**, **120**, **122** is fastened to the cargo box **50** using a plurality of screws, bolts, rivets or any other suitable fastener. In other embodiments, at least one of the track members **118**, **120**, **122** is directly welded to the side wall **52** of the cargo box **50**. In the delivery vehicle **10** of FIGS. **1-6**, each of the track members **118**, **120**, **122** is fastened to the cargo box **50** using a series of bolts **208** that interface with a corresponding one of a plurality of through-holes disposed centrally along the length of each of the track members **118**, **120**, **122**.

(33) As shown in FIG. **8**, the side door **114** is configured to be inserted (e.g., nested within) a recessed area **51** in the side wall **52** of the cargo box **50**. Among other benefits, this nested configuration reduces aerodynamic drag on the cab **40** while the delivery vehicle **10** is in motion and improves sealing between the side door **114** and the side access **200**. The recessed area **51** includes a sealing wall **53** (e.g., lip, flange, etc.) extending inwardly along a perimeter of the side access **200**. As shown in FIG. **8**, the sealing wall **53** is located proximate to an interior surface of the side door **114** when the side door **114** is in the closed position. In an exemplary embodiment, a gasket or other compliant material is disposed along a sealing surface of the sealing wall **53**, between the sealing wall **53** and the interior surface of the side door **114** to prevent particulate and water ingestion into the cargo box **50**. Other surfaces of the recessed area **53** may also be considered as sealing surfaces in that each faces a corresponding one of the interior surfaces of the sliding door, which prevents particulate and water ingestion into the cab **40**.

(34) As shown in FIGS. **10-11**, a first end of each of the track members **118**, **120**, **122** is curved inwardly (e.g., toward the storage volume of the cargo box **50** of FIGS. **1-2**). This configuration allows the side door **114** to move at least partially laterally inward toward the storage volume along the curved portion of each of the track members **118**, **120**, **122** such that, in the closed position, an outer surface of the side door **114** is approximately coplanar with an exterior surface of the side wall **52** (see FIG. **5**). As shown in FIGS. **10-11**, the side door **114** is slidably coupled to each of the track members **118**, **120**, **122** via connective sub-assemblies. A connective sub-assembly **124** for the upper track member **118** is shown in FIG. **10**. The connective sub-assembly **124** is configured to maintain the side door **114** in parallel alignment with the side wall **52** of the delivery vehicle **10** (e.g., to prevent the side door **114** from contacting the side wall **52**) when the side door **114** is reconfigured from the closed position to the open position (see also FIG. **1**). The connective sub-

assembly **124** includes a roller assembly having wheels or rollers that interface with one or more surfaces of a corresponding track member **118**, **120**, **122**. The roller assembly is pivotably coupled to (such that it may at least partially rotate and/or pivot relative to) a first end of an adapter that extends between the roller assembly and an interior surface of the side door **114**. The second end of the adapter is rigidly coupled to the interior surface of the side door **114**. In an embodiment, the adapter may be fastened to the interior surface of the side door **114** using bolts, screws, or any other suitable fastener. In other embodiments, the adapter may be welded to the interior surface of the side door **114**.

(35) Returning to FIG. **6**, the cargo box **50** includes a rear door **302** that is slidably engaged to a rear facing side wall **52** of the cargo box **50**. In various embodiments, the rear door **302** is a roll-up or coiling door that is configured to retract toward the upper wall **54** of the cargo box **50**. In other embodiments, the rear door **302** is configured as a sectional door (e.g., a garage-style door including multiple hingedly connected sections) that slides along two substantially parallel guide rails coupled to the cargo box **50**. In yet other embodiments, the rear door **302** is a canopy door.

(36) In the exemplary embodiment of FIGS. **1-6**, the delivery vehicle **10** includes a roof cap assembly, shown as roof cap **400**, disposed on an upper surface **42** of the cab **40**. As shown in FIGS. **1-6**, the roof cap **400** is rigidly coupled to both the upper surface **42** of the cab **40** and the forward partition **100** (e.g., a forward surface of the cargo box **50**) using a series of fasteners (e.g., bolts, screws, etc.). In other embodiments, the roof cap **400** may be welded to at least one of the upper surface **42** of the cab **40** and the forward partition **100**. As shown in FIGS. **3-4**, once installed, the roof cap **400** extends between an A-pillar **43** of the cab **40** (e.g., a support pillar forward of the cab **40** toward the front windshield) and a B-pillar **44** of the cab **40** (e.g., a vertically oriented support pillar toward the rear of the cab **40**). As shown in FIG. **5**, a width **401** of the roof cap **400** in a lateral direction (e.g., between the driver side and the passenger side of the delivery vehicle **10**) is approximately equal to a width **49** of the cab **40** and a width **55** of the cargo box **50** in order to reduce drag on the delivery vehicle **10** during operation.

(37) Referring now to FIG. **12**, the roof cap **400** (not shown) is configured to completely cover (e.g., enclose) a cutout **46** disposed in a top wall **48** of the cab **40**. Among other benefits, the cutout **46** provides vertical clearance to an occupant who may be required to access the storage volume of the cargo box **50** through partition access **110**. Advantageously, the roof cap **400** accommodates the transition in height between the cab **40** and the cargo box **50**. Various designs are contemplated for the roof cap **400**. In the exemplary embodiment of FIGS. **13-15**, the roof cap **400** is formed from one or more pieces of lightweight material (e.g., injection molded plastic, aluminum, etc.). The roof cap **400** includes a forward wall **402** and lateral walls **404** that are substantially perpendicular to the forward wall **402** that, together, define an enclosed interior space **406**. As shown in FIGS. **13-15**, the forward walls **402** of the roof cap **400** and the transition between the forward walls **402** and the lateral walls **404** are curved to reduce aerodynamic drag on the delivery vehicle **10** during operation. As shown in FIG. **14**, the roof cap **400** may further include one or more support members **408**, which interface with the upper surface **42** of the cab **40**.

(38) As shown in FIG. **14**, the roof cap **400** includes a mating interface, shown as lip **410** that extends along a perimeter of a rear portion of the roof cap **400**. A series of mounting holes **412** are disposed in the lip **410**. Each of the mounting holes **412** is alignable with a corresponding one of a plurality of receiving holes **414** (FIG. **12**) in the forward partition **100**. In other embodiments, the lip **410** may extend at least partially along a perimeter of a lower portion of the roof cap **400** or a perimeter of both the rear portion and the lower portion. The roof cap **400** is coupled to at least one of the top wall **48** (e.g., upper surface **42**) or the forward partition **100** (e.g., one or both of the top wall **48** and forward partition **100**) using a series of bolts, screws, rivets, or another suitable fastener, each of which pass through one of the mounting holes **412** and corresponding one of the receiving holes **414**. As shown in FIGS. **1-2**, an upper edge **403** of the roof cap **400** is approximately flush with the upper wall **54** such that the roof cap **400** is approximately co-planar

with the upper wall **54** proximate to the upper edge **403**. As shown in FIG. **15**, at least one surface of the roof cap **400** is contoured to match the shape of the upper surface **42** of the cab **40**, which, advantageously, reduces the size of the gap between the roof cap **400** and the upper surface **42**. In an embodiment, the roof cap **400** may further include a gasket or other sealing material (e.g., disposed along the lip **410**) to improve sealing between the cab **40** and the surrounding environment.

(39) Referring now to FIGS. **16-17**, the cargo box **50** includes a forward partition **100** proximate to a rear wall of the cab **40** (see FIG. **2**). The forward partition **100** separates an occupant of the delivery vehicle **10** (e.g., an interior of the cab **40**) from the storage volume (e.g., the interior of the cargo box **50**). In the event of a traffic accident or collision, the forward partition **100** prevents any loose cargo from escaping the cargo box **50** and injuring the occupant. In an exemplary embodiment, a first portion of the forward partition **100** is made from perforated sheet steel. In other embodiments, the first portion of the forward partition **100** may be made from solid sheet steel, aluminum, or various other materials. The sheet steel is coupled (e.g., welded, fastened, etc.) to a plurality of substantially vertical frame members, shown as frame members **126** that extend between the lower wall **56** and upper wall **54** of the cargo box **50**. In addition to supporting the forward partition **100**, the vertical frame members **126** may also support the cab **40**, which may be bolted, riveted or otherwise fastened to at least one of the forward partition **100** and vertical frame members **126**.

(40) As shown in FIGS. **16-17**, the occupant may access the storage volume via access **110**, which is a rectangular shaped cutout centrally disposed in the forward partition **100**. The access **110** is sized to accommodate a standing height (e.g., full vertical height) of an occupant of the delivery vehicle **10**. In other words, a height **109** of the access **110** is greater than a maximum height of the cab **40** (e.g., an interior volume of the cab **40**). As shown in FIGS. **18-19**, an access door **128** is disposed in access **110** of the forward partition **100**. The access door **128** is slidably engaged to the forward partition **100** by track assembly **130**. As shown in FIG. **19**, the track assembly **130** for the access door **128** includes at least one guide rail, shown as rail **132** that is rigidly coupled to at least one of the forward partition **100** and one or more vertical frame members **126**. The track assembly **130** additionally includes a mating rail, shown as door rail **134** that is rigidly coupled to the access door **128** (e.g., using bolts, screws, a weld joint, etc.). As shown in FIG. **19**, the door rail **134** is received within the rail **132** and slides along the rail **132** toward the side wall **52** of the cargo box **50** (e.g., in a lateral direction, in a direction that is substantially normal to the side wall **52** of the cargo box **50**). In an exemplary embodiment, the track assembly **130** is a telescoping slide rail. Alternatively, the track assembly **130** may include a connective sub-assembly similar to that used for the side access door assembly **112** of FIGS. **10-11** or another sliding mechanism that provides suitable structural support for the access door **128**.

(41) As shown in FIGS. **20-21**, the cargo box **50** further includes a frame including a plurality of cargo frame members, shown as frame members **58** disposed at regular intervals along the length (e.g., forward-to-backward) of the cargo box **50**. Each of the frame members **58** is disposed proximate to a side wall **52** of the cargo box **50** and is oriented in a direction that is substantially normal to a lower wall of the cargo box **50** (e.g., the cargo frame members extend upwardly from the lower wall). The frame members **58** are configured to reinforce the cargo box **50** and provide a mounting location for a plurality of shelves **500**.

(42) A plurality of shelves **500** is disposed in the storage volume of the cargo box **50** as shown in FIG. **21**. The shelves **500** are configured to accommodate packages and other cargo in the storage volume. As shown in FIGS. **21-22**, shelves **500** of various different sizes and shapes may be utilized. As shown in FIG. **23**, each of the shelves **500** includes a base wall **502** and two mounting legs **504**. Each of the mounting legs **504** is coupled to the base wall **502** proximate to an end of the base wall **502**. Each of the mounting legs **504** is oriented in a direction that is substantially perpendicular to the base wall **502**. Each of the shelves **500** may further include one or more

retaining elements, shown as retaining lips **510** that prevent packages and cargo from sliding off of the shelves **500** while the delivery vehicle **10** is in transit.

(43) As shown in FIG. **23**, each of the shelves **500** is disposed between two adjacent frame members **58** in the cargo box **50**. As shown in FIG. **24**, each of the shelves **500** is pivotably coupled to the frame members **58** using a bolt **506**, or another suitable fastener. Each of the shelves **500** is configured to rotate toward an upper wall of the cargo box **50** (e.g., a roof of the cargo box **50**). An angular position of each of the shelves **500** may be secured using a locating member (e.g., a pin, bolt, etc.), shown as pin **508**. As shown in FIG. **24**, the pin **508** engages with a through hole **514** in one of the mounting legs **504** and a corresponding through hole **516** in one of the frame members **58**. In some embodiments, the frame members **58** may each include multiple through holes **516** to secure the base wall **502** at different angular positions relative to one of the side walls of the cargo box **50**. As shown in FIG. **25**, a contact element, shown as contact pad **512** is rigidly coupled to at least one mounting leg **504** of each of the shelves **500** and may be used to set a nominal angular position of the shelves **500** for storing various items (e.g., a position that prevents any packages or cargo from dropping from the shelves **500**, a position of the shelves **500** under vertical loading). As shown in FIG. **25**, the contact pad **512** is engaged with one of the frame members **58**. Specifically, the contact pad **512** contacts a stop plate **518** that is rigidly coupled to the frame member **58** (e.g., via bolts or another suitable fastener). In some embodiments, the position of the stop plate **518** may be adjustable to support the base wall **502** at different angular positions relative to the side wall of the cargo box **50**. Among other benefits, using adjustable shelving **500** increases the utility of the cargo box **50**, which may be adapted to suit a variety of different storage configurations.

(44) As utilized herein, the terms “approximately”, “about”, “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

(45) It should be noted that the term “exemplary” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

(46) The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

(47) References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

(48) Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply

that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

(49) It is important to note that the construction and arrangement of the elements of the systems and methods as shown in the exemplary embodiments are illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claims.

Claims

1. A vehicle, comprising: a chassis; and a vehicle body coupled to the chassis, the vehicle body comprising: a cab disposed on a forward end of the chassis; and a cargo body disposed on a rear end of the chassis, the cargo body including: a forward partition that engages the cab, the forward partition defining an opening, a height of the opening being greater than a height of the cab; and an access door disposed in the opening, and slidably engaged to the forward partition.
2. The vehicle of claim 1, wherein the height of the opening is sized to accommodate a standing height of an occupant of the vehicle.
3. The vehicle of claim 1, wherein the cargo body further comprises: a lower wall; a plurality of frame members extending upwardly from the lower wall; and a plurality of shelves pivotably coupled to the plurality of frame members.
4. The vehicle of claim 1, wherein the cargo body further includes: a lower wall; a plurality of frame members extending upwardly from the lower wall; and a plurality of shelves pivotably coupled to the plurality of frame members, wherein each one of the plurality of shelves comprises a base wall and a locating member configured to set an angular position of the base wall relative to one of the plurality of frame members.
5. The vehicle of claim 1, further comprising: a plurality of frame members coupled to the cargo body; and a shelf pivotably coupled to at least one of the plurality of frame members and movable between a folded position in which the shelf is retracted toward a side wall of the cargo body and an extended position in which the shelf extends away from the side wall, the shelf disposed between adjacent ones of the plurality of frame members, wherein in the folded position the shelf is retracted into a space formed between adjacent ones of the plurality of frame members.
6. The vehicle of claim 5, wherein the shelf comprises a base wall and a locating member configured to set an angular position of the base wall relative to at least one of the plurality of frame members.
7. The vehicle of claim 1, wherein the vehicle body further comprises a roof cap assembly extending between an upper surface of the cab and a forward surface of the cargo body.
8. The vehicle of claim 7, wherein a width of the roof cap assembly is approximately equal to a width of the cab, and wherein the width of the cab is approximately equal to a width of the cargo body.
9. The vehicle of claim 1, wherein the height of the cab is a distance between a floor of the cab and

an upper end of a B-pillar of the cab.
10. The vehicle of claim 1, wherein the height of the opening is sized to accommodate a standing height of an occupant of the vehicle.
11. The vehicle of claim 1, further comprising: a plurality of frame members disposed within the cargo body; a shelf pivotably coupled to at least one of the plurality of frame members and movable between a folded position in which the shelf is retracted toward a side wall of the cargo body and an extended position in which the shelf extends away from the side wall, the shelf including: a base wall and a locating member configured to set an angular position of the base wall relative to at least one of the plurality of frame members; and a mounting leg coupled to the base wall and oriented substantially perpendicular to the base wall, wherein the locating member engages the mounting leg.